

Artificial Intelligence

Introduction to Artificial Intelligence

Lecture 01 and Introduction

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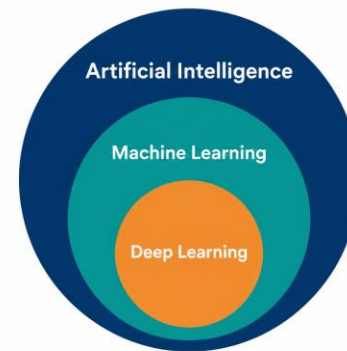


Lecture overview



This lecture develops introduction to artificial intelligence from its basic definition to its mathematical operation and practical use. The explanation follows a gradual sequence so that each new idea connects with the previous one.

The main discussion covers Artificial Intelligence, Human and artificial intelligence, Uses and value of artificial intelligence, Applications of AI. Numerical or practical examples show how the concepts produce observable outputs.



Example of the learning sequence: identify the inputs, apply the stated rule or equation, calculate the output, and interpret the result within the limits of the model.



AI studies systems that perform tasks associated with human intelligence. Typical capabilities include learning, reasoning, perception, language, and decision making. For example, classify a new image as Cat or Dog. An AI system converts observations into an internal representation, applies a model or a set of rules, and produces an action or prediction. Its apparent intelligence remains limited by the task, data, and evaluation method defined by its designers.

An image classifier receives pixel values and returns $P(\text{Cat}) = 0.92$ and $P(\text{Dog}) = 0.08$. The higher probability determines the predicted class, while a new test set measures reliability.

Human and artificial intelligence



Humans learn from experience and use broad context. AI systems learn from data or rules and operate within a defined task. Performance depends on data quality, model design, and evaluation. AI can support human decisions without replacing human responsibility. An AI system converts observations into an internal representation, applies a model or a set of rules, and produces an action or prediction. Its apparent intelligence remains limited by the task, data, and evaluation method defined by its designers.

A medical model can highlight a suspicious region in an X-ray within seconds. A physician then checks the image, the patient history, and possible causes before making a clinical decision.

Uses and value of artificial intelligence



Large datasets exceed manual analysis capacity. Complex patterns may resist fixed rules. Trained models can make repeated predictions quickly. Automation can improve consistency when the task is well defined. An AI system converts observations into an internal representation, applies a model or a set of rules, and produces an action or prediction. Its apparent intelligence remains limited by the task, data, and evaluation method defined by its designers.

A 3 by 3 letter becomes a nine-value bipolar vector. If one pixel is reversed, the similarity score may decrease but can still retain the correct sign and class.

Applications of AI and Major areas of AI



Healthcare: image analysis and risk prediction. Industry: inspection and predictive maintenance. Banking: fraud detection. Communication: speech recognition and translation. Education: adaptive learning. E-commerce: recommendation systems. Machine Learning learns patterns from data. Natural Language Processing works with human language. Computer Vision interprets images and video. Robotics connects sensing, planning, and action. Expert Systems apply encoded domain knowledge. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

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Traditional programming



The programmer supplies data and explicit rules. The program applies the rules to produce an output. This approach works well when the logic is stable and fully known. Data and Rules produce Output. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

A fixed rule can label a student as Pass when grade ≥ 50 . Machine learning is more suitable when success depends on several interacting features whose relationship is not known in advance.

A short history of AI



1950: Alan Turing proposed a test for machine intelligence. 1956: the Dartmouth workshop established AI as a research field. Later progress combined larger datasets, faster computers, and improved algorithms. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

A small numerical test should use known inputs and an expected output. Comparing the computed result with the expected result reveals whether the method and its assumptions were applied correctly.

Goals of artificial intelligence and Intelligent agents



Represent knowledge in a form a machine can use. Reason about facts, rules, and uncertain evidence. Select actions that achieve a stated objective. Learn from data or interaction. An agent receives information through sensors. It changes the environment through actuators. A rational agent selects the action expected to produce the best result. An AI system converts observations into an internal representation, applies a model or a set of rules, and produces an action or prediction. Its apparent intelligence remains limited by the task, data, and evaluation method defined by its designers.

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Problem solving in AI



Define the initial state and the goal state. List the allowed actions and their costs. Search for an action sequence that reaches the goal. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

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Knowledge representation



Facts describe objects and their properties. Rules connect conditions with conclusions. Graphs represent relationships among concepts. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

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Artificial intelligence and human intelligence and AI benefits



Artificial intelligence Fast calculation and consistent repetition. Learns within the data and objective provided. Can operate continuously. Human intelligence Uses broad context and common sense. Transfers knowledge between unfamiliar tasks. Combines reasoning with emotions and social understanding. Automates repetitive analysis. Supports decisions using large datasets. Improves accessibility through speech and vision tools. Detects patterns that may be difficult to notice manually. An AI system converts observations into an internal representation, applies a model or a set of rules, and produces an action or prediction. Its apparent intelligence remains limited by the task, data, and evaluation method defined by its designers.

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AI risks and limitations



Biased data can produce biased predictions. Complex models may be difficult to explain. Incorrect outputs can cause harm when oversight is weak. Systems require security, privacy, and responsible use. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

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route planning



State: the current city and road network. Actions: travel through a connected road. Cost: distance or travel time. Goal: reach the destination with minimum cost. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

A route planner compares A to B with cost 5 against A to C to B with costs 2 and 1. The second route has total cost 3, so the search algorithm selects it.

Responsible use discussion



Tasks that require intelligence can be distinguished from fixed automation by their need for adaptation, reasoning, or uncertain decisions. Evidence of reliability comes from representative testing, error analysis, and monitoring after deployment. Human responsibility remains essential when decisions affect safety, rights, or major resources. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

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Integrated application



A complete application of introduction to artificial intelligence begins with a clearly defined input and target. The method applies its architecture, learning rule, or processing equation to obtain a measurable output. Evaluation compares this output with the expected response, quantifies the error, and checks whether the model generalizes beyond the training examples. The interpretation should also consider the assumptions of the algorithm and the quality of the input data.

A small numerical test should use known inputs and an expected output. Comparing the computed result with the expected result reveals whether the method and its assumptions were applied correctly.

Lecture summary



The lecture connected artificial intelligence, human and artificial intelligence, uses and value of artificial intelligence with applications of ai, major areas of ai, traditional programming. These concepts describe how an intelligent or neural system represents inputs, performs a calculation, learns from error or association, and produces a result.

The central principle is that a model should be understood through its inputs, parameters, mathematical operation, output, and evaluation method. Each part affects the reliability of the final result.

Example connection: a correct calculation is useful only when the data and assumptions represent the real task.